

Appearance-Based Object Recognition Using Weighted Longest Increasing Subsequence

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Abstract

We proposed in this paper a novel weighted longest increasing subsequence to improve the performance of the appearance-based object recognition. The LIS is employed to find the true keypoint matches that have consistent geometric order in both query and gallery images. Then, the similarity between query and gallery images is measured by the sum of the weights of the true keypoints. The experimental results shown that our approach outperforms the SURF and SURF + RANSAC Homography approaches.

1. Introduction

The rise of the camera-phones and the popularity of the smart-phones have triggered a huge demand of object recognition related applications, such as landmark recognition, learning trail, advertisement campaign, and shop on the go. Appearance-based object recognition is the core technology that provides such services. It recognizes a 3D object using a snap of its image, a 2D view of the object. The appearance may vary between captures due to illumination, viewpoint and scaling variations. It also affected by occlusion and imaging noises. All these variations make the appearance-based object recognition a challenging task.

Mikolajczyk and Schmid [8] evaluated the performance of various local descriptors for object recognition under different geometric and photometric transformations. It was shown that the SIFT-based features, Scale-Invariant Feature Transform (SIFT) [7] and Gradient Location and Orientation Histogram (GLOH) [8], achieved the best performance. They are robust to common image variations such as image scale, viewpoint and illumination changes. Although the feature match-

ing approaches may provide relatively good recognition performances, they still suffer from false keypoint matches (outliers). Therefore, there are typically one or more follow-up verification steps to verify the keypoint matches.

Geometric verification approaches such as RANSAC Homography [6] have been proposed in the literature to solve the outlier problem. The RANSAC Homography uses the Random Sample Consensus (RANSAC) [4] to find the best homography model for a set of keypoint pairs and rejects all keypoint matches that are not conforming to the found homography. Although this method works well in many applications, they perform poorly when the number of false keypoint matches outnumbers the number of true keypoint matches or when the number of keypoint matches is limited.

In this contribution, we propose a weighted longest increasing subsequence (WLIS) approach for appearance-based object recognition. It is a novel geometric verification method that defines the true keypoint matches as a subset of consistent corresponding keypoints. A subset of matched keypoints is consistent if the geometric order of the corresponding keypoints is the same in both query and gallery images. We find the largest subset of corresponding keypoints by calculating the Longest Increasing Subsequence (LIS) [5].

The similarity between query and gallery images is then measured by the weighted sum of the largest subset of the consistent corresponding keypoints. Whereas, the weight is defined by one minus the ratio between the Euclidean distances of the best and second-best matched descriptors. Based on our experiments, it is shown that our WLIS geometric verification approach performs better than the RANSAC Homography [6]. Also, it run faster and required less computational load than the RANSAC Homography.

The rest of this paper is organized as follows. In

Section 2, our appearance-based object recognition approach using the weighted longest increasing subsequence is presented. Section 3 describes the experiments used to evaluate the performance of the proposed approach. Finally, we conclude in Section 4 with an overview of the main results.

2. Appearance-Based Object Recognition

This section describes our appearance-based object recognition approach. It consists of a set of processes: feature detection and matching, weight calculation, geometric verification, and image similarity measure.

2.1. Feature Matching and Weight Calculation

Object recognition is performed by matching a query image to the gallery images of the objects. Local features or keypoints are detected from the images and matched based on their descriptors. Here, we adopt the Fast-Hessian keypoints detector and Speeded Up Robust Features (SURF) descriptor proposed by Bay et al. [1] because of its speed and robustness. The dissimilarity between two features is measured by the Euclidean distance of their descriptors. The best matches between features in the query and gallery images are defined by the k-d tree data structure and search algorithm [2].

In the original implementation of the SIFT [7] or SURF [1] matching for object recognition, a pair of features is considered a match if the distance ratio between the best and the second-best matches is below a certain threshold. Even though this approach can remove some of the false matches, however the selection of suitable threshold value may not be trivial. Therefore, we avoid performing such thresholding in this early stage. Instead, we provide a weight for every pair of best matches as a measure of the quality of their match. The query features are matched to the features of all images in the gallery. The weight of the i^{th} feature of the query image (w_i) is defined as:

$$w_i = 1 - d_1/d_2, \quad (1)$$

where d_1 is the distance to the best matched feature in the gallery image and d_2 is the distance to the second best matched feature. The second-best matched feature has to come from a different class. Since $d_1 \leq d_2$ and $d_1, d_2 \geq 0$, then the range of the weight values is from 0 to 1.

2.2. Longest Increasing Subsequence

The matched features or keypoint pairs are subjected to a geometric verification. We propose a novel geometric verification approach that finds the true keypoint matches. The true keypoint pairs are a subset of the corresponding keypoints that have the same geometric order in both query and gallery images. We find the largest true keypoint matches by calculating the LIS [5].

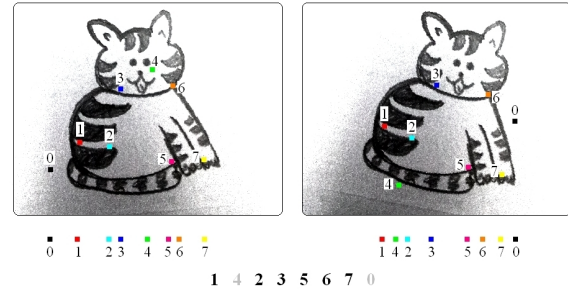


Figure 1. An illustration of finding the Longest Increasing Subsequence on two images based on the X-coordinates of the detected keypoint pairs. The LIS output in this case is 1, 2, 3, 5, 6 and 7

Figure 1 illustrates an example of finding the LIS of the keypoint pairs along the X-axis. Here, the corresponding keypoints on the two images are labeled by the same number. First, we sort the keypoints on both images separately according to their X-coordinates. Then, we keep the largest subset of the keypoint pairs that has a consistent sequence on both images. This LIS along a single axis is invariant to translation and scaling. The geometric order of the keypoints does not change when we translate or scale the images. However, it is not invariant to rotation.

Computing the LIS is very fast. We can compute it multiple times in order to simulate rotations. We randomly rotate the keypoints of the query and train images and re-compute the LIS. If we do it enough times, we will have a high chance of finding the best axis, along which we can obtain the largest geometrically consistent subset of keypoints. Based on our experiments, we recommend computing LIS on three equally spaced angles for the query image and seven for the gallery image.

The proposed LIS approach is invariant to affine transformations. Let $Affine : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an affine transformation. $Affine(\vec{p}) = \mathbf{A}\vec{p} + \vec{t}$ where $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the affine transformation matrix, and

$\vec{t} = \begin{bmatrix} e \\ f \end{bmatrix}$ is the translation vector. Let $Rotate : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a rotation. $Rotate(\vec{p}) = \mathbf{R}\vec{p}$ where $\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and θ is the angle of the rotation. If we apply an affine transformation on a point and also apply the rotation, we will get $\vec{p}' = Rotate(Affine(\vec{p}))$. This can be expressed as:

$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \times \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix} \right), \quad (2)$$

where $\vec{p}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}$ and $\vec{p}'_i = \begin{bmatrix} x'_i \\ y'_i \end{bmatrix}$. Since the LIS is performed along a single axis, without loss of generality, we can reduce the problem to the sequence of keypoint pairs along the X-coordinates. The difference between the X-coordinates between two keypoints after transformation can be defined as:

$$x'_1 - x'_0 = (a \cdot \cos \theta - c \cdot \sin \theta) \cdot (x_1 - x_0) + (b \cdot \cos \theta - d \cdot \sin \theta) \cdot (y_1 - y_0) \quad (3)$$

In practice we do not know θ , because we do not know the parameters of the affine transformation. Given that we obtain the “right” angle from the simulated rotations, as described above, so we can define $b = r \cdot \sin \theta$ and $d = r \cdot \cos \theta$, where $\theta, r \in \mathbb{R}^2$ and $r > 0$. After substitution, Eq. (3) can be rewritten as:

$$x'_1 - x'_0 = \left(\frac{ad - bc}{r} \right) \cdot (x_1 - x_0). \quad (4)$$

Since $\left(\frac{ad - bc}{r} \right) = \frac{\det(\mathbf{A})}{r} > 0$, therefore we can preserve the relative order of keypoints in the affine transformed image by applying a carefully chosen rotation.

2.3. Image Similarity Measure

After LIS verification, we only keep true keypoint matches that are consistent with the geometrical constrain. The similarity between query and gallery images is then defined by the sum of the weights of the remaining keypoints. The similarity score (S) is defined as:

$$S = \sum_i w_i f_i, \quad (5)$$

where f_i is equal to 1 for a consistent feature or 0 for otherwise. This similarity score is used to judge whether a pair of query and gallery images belong to the same class or object according to a selected threshold value.

3. Experiments

We evaluate the performance of our approach on our advertisement and landmark database as well as on the Stanford Mobile Visual Search (SMVS) database [3]. This section describes the datasets used in the experiments, methodology, and experimental results.

3.1. Datasets

The experiments are performed on nine separate datasets from two different databases. Two of the datasets are from our advertisement and landmark database, while seven datasets are from the SMVS database. Our advertisement dataset contains 10810 images and landmark dataset contains 4793 images. We also collect 26604 negative images that are not belonging to any of the classes in the datasets. We randomly select 1300 negative images to be used as negative samples for the datasets of SMVS database. The number of classes, gallery images, query images, and negative images for each dataset are summarized in Table 1.

Table 1. Summary of images allocation and number of classes for different datasets

Datasets	No. of Classes	No. of Gallery Images	No. of Query Images	No. of Negative Images
Advertisements	1201	1201	9609	26604
Landmarks	130	3033	1760	26604
Book Covers	101	101	404	1300
Business Cards	100	100	400	1300
CD Covers	100	100	400	1300
DVD Covers	100	100	400	1300
Paintings	91	91	364	1300
Text Documents	100	100	400	1300
Video Frames	100	100	400	1300

3.2. Methodology

The similarity score between query and gallery images is calculated based on the method described in Section 2.3. The gallery image with the highest similarity score is selected as the matched image. If the query image and the matched image belong to the same class, then their score will be treated as a correct score. Otherwise, it will be treated as an incorrect score. The negative images is also matched to the gallery images. The

similarity score between negative and gallery images is treated as a negative score.

The performance of an approach is measured by a performance value. A higher performance value indicates a better object recognition approach. The performance value (E) is defined as

$$E = (CRR \times CJR) / (1 + IRR) \quad (6)$$

The correct recognition rate (CRR) is defined as the ratio between the correct scores and the number of query images. While, the incorrect recognition rate (IRR) is defined as the ratio between the incorrect scores and the number of query images. These CRR and IRR vary according to the selected threshold as some of the correct and incorrect scores are rejected if their scores are lower than the threshold. The correct rejection rate (CJR) is then defined as ratio of the negative scores that are less than the threshold. The threshold is selected such that to maximize the performance value (E).

3.3. Experimental Results

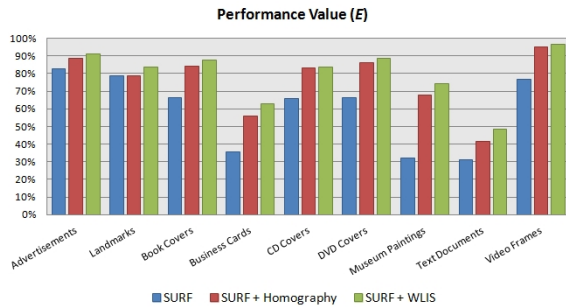


Figure 2. Bar graph of performance values of the evaluated approaches on different datasets

We compare the performance of our approach, the SURF + WLIS, with two other approaches, the SURF matching [1] and the SURF + RANSAC Homography [6]. Their maximum performance values on different datasets are summarized in Figure 2. On average, the proposed approach achieved 20% increase in performance value compared to the SURF matching and 4% compared to the SURF + RANSAC Homography.

Table 2 summarizes the processing speed of the approaches based on the advertisement and landmark datasets. It shows the processing speed of the WLIS is faster than the RANSAC Homography. The average query time is measured on an Intel XeonTM 5150 CPU

Table 2. Processing speed of the evaluated approaches on the advertisement and landmark datasets

Datasets	Average Query Time (ms)		
	SURF	SURF + Homography	SURF + WLIS
Advertisements	22.5	55.3	37.7
Landmarks	53.2	73.6	72.1

having a 3.00 GHz processor and 3.00 GB RAM with an operating system of WindowsTM XP SP3.

4. Conclusion

We proposed in this paper a novel weighted longest increasing subsequence to improve the performance of the appearance-based object recognition. Experimental results based on our database and the SMVS database have been presented. There are 45,667 sample images with challenging variations in both databases. The proposed approach outperforms the SURF matching and SURF + RANSAC Homography approaches. The experimental results have demonstrated the validity and effectiveness of our approach.

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